Chapter 1 Riemannian Metrics

1.1 Definitions

Definition 1.1.0.1

- Let M be a smooth manifold. A **Riemannian metric** on M is a 2-tensor field $g \in \mathcal{T}^2 M$, whose value

$$g_p : T_pM \times T_pM \to \mathbb{R}$$

  which is positive definite in the sense of $g_p(v, v) \geq 0$.

- A **Riemannian manifold** is a pair (M, g).

Remark Riemannian metric vs Metric Space

♣ **Example 1.1.0.2** (Eucilidean Metric)

Isometires

Local Representation

Definition 1.1.0.3 (Local Representation of Riemannian Metric)

♠ **Proposition 1.1.0.4** (Existence of Orthonormal Frame)

∘ **Proof** ... □

Definition 1.1.0.5 (Unit Tangent Bundle) Define the **unit tangent bundles**

$$UTM := \{(p, v) || v|_g = 1\}$$

♠ **Proposition 1.1.0.6** If (M, g) is a Riemannian-Manifold with or without boundary, it unit tangent bundle UTM is a smooth, properly embedded codimensional-1 submanifold with boundary in TM, with

$$\pi : UTM \to M \qquad \partial(UTM) = \pi^{-1}(\partial M)$$

Then, UTM is connected⇔ M is connected. UTM is compact⇔ M is compact.

∘ **Proof** ... □

1.1.1 Riemannian Submanifold

◇ **Lemma 1.1.1.1** Suppose $(\tilde{M}, \tilde{g})$ is a Riemannian manifold with or without boundary. M is a smooth manifold with or without boundary. And

$$F : M \to \tilde{M}$$

is a smooth map. The smooth 2-tensor field(pullback)

$$g = F^* \tilde{g}$$

is a Riemannian metric iff F is an immersion.

∘ **Proof** ... □

1.2 Models of Riemannian Manifolds

1.2.1 Symmetries

1.2.2 Covering Space of Riemannian Manifold

Eucilidean Space

Sphere

Hyperbolic Space

Killing-Hopf Theorem

▶ ***Theorem*** Suppose (M, g) be compelete, simply connected Riemannian n-manifold with constant sectional curvature, $n \geq 2$.
Then, M is isometric to one of the model spaces $\mathbb{R}^n, \mathbb{S}^n(R), \mathbb{H}^n(R)$.

∘ **Proof** ... □

★ **Corollary 1.2.2.1** ...

∘ **Proof** ... □

Space Form

Chapter 2 Riemannian Connections

2.1 Affine Connections

Definition 2.1.0.1 Define the **affine connections**

$$\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M)$$
$$(X, Y) \mapsto$$

With following properties:

- ($C^\infty(M)$-Linear) $\nabla_{fX+gY} Z = f\nabla_X Z + g\nabla_Y Z$;
- ($\mathbb{R}$-Linear) $\nabla_X(aY + bZ) = a\nabla_X(Y) + b\nabla_X(Z)$
- (Lebnitz)$\nabla_X(fY) = f\nabla_X Y + X(f)Y$

where $X, Y, Z \in \mathfrak{X}(M)$, $f, g \in C^\infty(M)$.

♠ **Proposition 2.1.0.2** Let M be a differentiable manifold with an affine connection ∇. There exists a unique correspondnece which associates to a vector field V along the differentiable curve

$$c : I \to M$$

another vector field aloing c, called the **covariant derivative** of V along c, such that:

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-

∘ **Proof** ... □

2.1.1 Parallel Vector Field

2.2 Riemannian Metrics, Levi-Civita Theorems

Chapter 3 Geodesics

3.1 The Geodesics Flow

3.2 Minimalizing Properties

Chapter 4 Curvatures

4.1 Curvatures

4.2 Sectional Curvature

4.3 Ricci Curvature and Scalar

4.4 Tensors on Riemannian Manifolds

4.5 Manifolds of Nonpositive Curvature

4.5.1 Cartan-Hadamard Manifold

▶ ***Theorem*** (Cartan-Hadamard)

∘ **Proof** ... □

Definition 4.5.1.1 ...

♠ **Proposition 4.5.1.2** ...

∘ **Proof** ... □

♠ **Proposition 4.5.1.3** ...

∘ **Proof** ... □

4.5.2 Cartan's Fixed Point Theorem

4.5.3 Cartan's Torsion Theorem

Chapter 5 Appendix: Curvatures by Curves

5.1 Classification of Curves: A Review

Definition 5.1.0.1 Given a C^1-curve $c : [a, b] \to \mathbb{R}^n$:

- c is called **regular** if

$$\|\dot{c}(t)\| \neq 0, t \in [a, b]$$

- c is called **path-parametrized** if $\|\dot{c}(t)\| = 1, t \in [a, b]$;
- c is called **closed**, if $c(a) = c(b)$;
- c is called **differentiable closed** if $c(a) = c(b); c'(a) = c'(b)$;

♠ **Proposition 5.1.0.2** Suppose $c : [a, b] \to \mathbb{R}^n$ be C^k-regular, then there exists a C^k-diffeomorphism

$$\mathrm{Diff}^k([0, 1], [a, b]) \ni \varphi : [0, l] \to [a, b] \qquad \tilde{c} = c \circ \varphi$$

where $\tilde{c}$ is path-parametrized, $\|\dot{\tilde{c}}\| = 1$;

∘ **Proof** ... □

Definition 5.1.0.3

- Given a regular C^1-curve $c : I \to \mathbb{R}^n$, then

$$T(t) := \frac{\dot{c}(t)}{\|\dot{c}(t)\|}$$

  is called the unit tangential vector field.

- Given a regular C^2-curve c. Then, define

$$\kappa(t) := \frac{\|T'(t)\|}{\|\dot{c}(t)\|}$$

  is called the **curvature** of c at t;

As a matter of fact, $T_t c : \mathbb{R} \cong T_t I \to T_{c(t)}\mathbb{R}^n \cong \mathbb{R}^n$, this is a linear map, then

$$T(t) = T_t c(1)$$

♠ **Proposition 5.1.0.4** Suppose c be regular. Then,

$$\kappa(t) = \frac{\|(\ddot{c}(t))^{\perp}\|}{\|\dot{c}(t)\|^2}$$

where $(\ddot{c}(t))^{\perp}$ is the orthogonal projection onto hyperplane $T(t)^{\perp}$, i.e. $= \ddot{c}(t) - T(t) < T(t), \ddot{c}(t) >$;

∘ **Proof** ... □

★ **Corollary 5.1.0.5** Suppose c be path-parametrized, then $\kappa(t) = \|\ddot{c}(t)\|$.

∘ **Proof** Boit □

♠ **Proposition 5.1.0.6** (Curvature is not related to Parametrimization)

∘ **Proof** ... □

Definition 5.1.0.7 Given a regular C^2 curve, with $\kappa(t_0) \neq 0, t_0 \in I$:

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5.2 Curvature of Curve

5.2.1 Total Curvature

◇ **Lemma 5.2.1.1** ...

∘ **Proof** ... □

5.2.2 Curvature on Plane

♠ **Proposition 5.2.2.1** We could calculating the unit tangent vector field and normal vector field as following

Definition 5.2.2.2 Define the spin operation

$$\begin{pmatrix} cos\theta & -sin\theta \\ sin\theta & cos\theta \end{pmatrix} \qquad J := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \theta = \frac{\pi}{2}$$

Then,

$$N(t) = JT(t)$$

∘ **Proof** □

And the curvature on plane is 'orientable', because of the property of $\mathbb{R}^2$:

5.2.3 Knotted Curves

5.3 Isoperimetric inequality

◇ **Lemma 5.3.0.1** Suppose $c : I \to \mathbb{R}^2$ be: simply closed, positive oriented, regular C^1-curve, and with coordinate

$$c(t) = (x(t), y(t))$$

Then, for the area described by the curve,

$$A = -\int_0^l x'(t)y(t) = \int_0^l x(t)y'(t)dt$$

∘ **Proof** ... □

♠ **Proposition 5.3.0.2** (Isometric Inequality)

Chapter 6 Riemannian Submanifold

Chapter 7 Gauss-Bonnet Theorem

Chapter 8 Jacobi Field

Chapter 9 Comparison Theory

Chapter 10 Curvature and Topology

Chapter 11 Isomorphic Immersions

Chapter 12 Compelete Manifold, Hopf-Rinow and Hadamard Theorems

Chapter 13 Spaces of Constant Curvature

Chapter 14 Variations of Energy

Chapter 15 The Morse Index Theorem

Chapter 16 The Fundamental Group of Manifolds of Negative Curvatures

Chapter 17 The Sphere Theorem

Chapter 18 The Laplacian

This will be given in geometric analysis.